

DANA 4800
Fall 2023
TERM PROJECT INFORMATION

The objective of the term project is to provide you with the opportunity to gain experience in study design, data collection, analysis and summary of data, contextual interpretation of data, and drawing conclusions from data. Statistical methods should be utilized in your term project. Students are encouraged to design and conduct a statistical investigation focusing on their particular topic of interest.

Each student is required to complete a term project. It will be worth 25 percent of your final grade. Those who fail to complete a satisfactory term project will receive a final grade for the course no higher than "D".

Students work in a group of four students whenever possible (if the number of students in a section is divisible by 4). Each team member in the project group is expected to contribute equally. The powerpoint file to be submitted should allocate one slide to include the names of team members and their estimated contribution in percentages. Members of any group who do not contribute to the term project in a satisfactory manner will receive a final grade for the course no higher than "D".

The requirements for successful completion of the term project are summarized as follows:

- Decide among team members on suitable topics. You are encouraged to discuss your ideas with your instructor.
- A project proposal (one page) outlining the objectives of your study, data source, and a description of variables will be due on **21st October, 2023**. You are welcome to hand in your proposal earlier if you wish to start the project sooner.
- You are to search for an (must be one) existing public dataset from reputable sources (one such source is www.kaggle.com). The dataset must have at least 150 rows of observations. You are required to provide the data files as part of your completed project.
- The **Presentation** will be in class on **28th and 30th November, 2023**. The powerpoint file should be uploaded to Brightspace by on November 26, 2023 by 11:59 pm. You should include an introduction, objectives of the study, literature review, methods, results (tabular and graphical summaries), summary of results, limitations of the study, conclusions and recommendations. The grading rubric for your final project is found on the last page of this document.

DANA 4800 - Fall 2023 Term project

The following are the items you should have in your final project.

I. Proposal [20 marks]

You should already have submitted your TCPS tutorial completion certificate before working on the proposal. A project proposal (1 page, pdf format only) is due on **October 21** (the proposal must be uploaded to Brightspace group folder (see Assessment->Assignment) before 11:59 pm on the date due). Each team submits ONE proposal. The proposal needs to outline the study objectives of your project, population of interest, information on how and where you obtain your dataset, sample size and a description of variables.

Your proposal must be

- Typed
- **Exactly** 1 page in length (any longer than 1 page may result in loss of marks)
- Your data collection involves using an existing publicly available dataset from a reputable source. Your proposal must identify the source of the dataset, including identifying which variables you will use (6 variables, 3 of which are quantitative variables, and 3 of which are categorical ones). You may get some ideas on suitable datasets from these links:
<https://github.com/awesomedata/awesome-public-datasets>
<https://www.kaggle.com/datasets>
<https://archive.ics.uci.edu/ml/datasets.php>

(you are not limited to what is listed here and you may seek your own datasets based on your team's interest).

In your proposal, you are to include

- Group members' names (student ID is optional, as a student has a right to keep his/her student ID private)
- Population
- Clear statements of the study objectives (can be phrased as questions) – e.g., Is there a linear relationship between two quantitative variables; is there an association between two categorical variables, what is the sample mean age, what is the sample proportion of people using Samsung phones etc. There should be exactly 5 statements (questions of interest), two for univariate analyses and three for bivariate analysis (see project details). The purpose of identifying these five questions of interest is so that you can provide answers to them in your project, providing a clear direction of what you are supposed to do in this project. Providing answers to these 5 questions is just a part of the project, and you are expected to do more than that (including drawing suitable graphs and charts, and applying as much of what you learnt in class to this project).
- Variables (you should have at least 6 variables – at least three categorical and at least three quantitative variables. You have to read the rest of this project document carefully in order to come up with 6 relevant variables. For example, two of the three quantitative variables should be carefully selected such that they might plausibly exhibit a linear relationship.)
- Identify which two variables are used for correlation/regression analysis
- Identify which two variables are used for chi-square analysis

- Sample size (the original dataset must have at least 150 rows).
- Source of dataset (where did you get this dataset).

The proposal is worth 20 marks (20% of the project) and marks are allocated as described below.

1. (4 marks) Define the questions of interest (that is, statements of study objectives).

These questions involve one or more variables. E.g. "Is there an association between gender and whether they smoke or not?" is a question involving two categorical variables. "What is the sample proportion of people who use Samsung phones?" is a question involving a single variable. There should be 5 questions of interest since you have to perform 3 univariate analyses and 2 bivariate analyses (regression, chi-square). Ideally, the questions are all related to a common topic (e.g., about smoking) so that the entire project appears connected and has a theme. The intention of these questions is so that you can find answers to these question through data analysis and address them in your powerpoint file (to be handed in). It takes time to find a suitable dataset so make sure you do not work on this at the last minute.

2. (8 mark) Define the population of interest, size

You must be specific. E.g., "students" is not specific enough for a definition of a population. Use the format as taught in class: "The population is made up of all" is a good definition.

3. (8 marks) Define the variables you use in your project and provide the source of your dataset.

Identify 6 variables, 3 of which are quantitative and 3 of which are categorical. When defining a quantitative variable that has a unit, include it in your definition. E.g., Weekly wage (in \$). Try to select variables that have a good chance of being strongly related to the question you are trying to answer. E.g., in regression, for every one-unit increase in the explanatory variable, the predicted value of the response variable changes by a fixed number of units. With that in mind, you should select two quantitative variables that have a good chance of being linearly related. Students should be aware that part of the project is to perform correlation **and** regression analysis, as well as a chi-square analysis. Much thought should be put into choosing suitable variables. Students have to identify which two variables are used for correlation/regression analysis, and which two variables are chosen for chi-square analysis. **(Marks are given for clarity of description, identifying type (categorical/quantitative), and choice of variables suitable for regression analysis and chi-square test).**

Please read the rest of this document before working on the proposal and make sure you understand the full project requirements in order to come up with a good proposal.

Details on the project:

Your powerpoint slides and data file must be uploaded to D2L/Brightspace. The powerpoint slides should cover the following:

Data Collection

1. Describe your data set.

2. How you obtained your data.

The data file should be provided (include units of measurement). You should upload the dataset.

3. Discuss possible bias.

Discuss the type(s) of bias you may expect in your study (even though you did not perform data collection but used a publicly available dataset).

Literature Review

You are to review 1 paper that use your intended statistical methods in analyzing similar data. Review should be a brief description of what was done to answer research questions in the paper.

Analysis

1. Three Univariate Data Analyses.

Analyze three of your variables (one of them must be categorical and one of them is quantitative, and the third one could be either quantitative/categorical). For quantitative variables, include the most appropriate graph, the most appropriate set of summary statistic (e.g., measures of centre and spread) and necessary R output. For categorical variables, include the most appropriate graph, the most appropriate statistic and necessary R output. You are to extract and include useful information from R output to include in your powerpoint file (do not include R output that you do not understand or where the information is not discussed in your analyses).

2. Two Bivariate Data Analyses

You should perform **(exactly) one correlation+regression analysis and one chi-square analysis.** Be sure to include suitable plots/graphs. *NOTE: Be sure to check the conditions for chi-square-analysis (all expected counts are at least 5). Extract only the information from R that you wish to include (do not simply copy and paste). No hand calculations are allowed and only R output are required, along with your interpretations of the results.*

It is my expectation that you apply as much of what is taught in class as possible. Go through the lecture notes and check that you did not miss anything that could be added.

Conclusion

What's the conclusion to the 5 questions you posed? Be sure to back up your answer by referring to your analysis. Summarize the results of your study. You also need to explain how you would improve this project if you were to do it again.

- Your completed project is in the form of a powerpoint presentation and must strictly be no longer than 15 minutes.
- This is a **group project**. The groups will consist of mostly 4 students (but may allow 5 under certain circumstances).
- Each team member in the project is expected to contribute equally and you should list down in one slide the names of the team members and their estimated contribution in percentages (e.g., Alex 25%; Bob 25%; Claire 25%; Donnie 25%)
- **Your project is graded as follows:**

Proposal	20 marks
Powerpoint content Part I Sampling and Data collection	10 marks
Literature Review	10 marks
Part II Analysis of univariate	20 marks
Analysis of bivariate	20 marks
Part III Conclusion	10 marks
Multimedia (relevant photos, videos etc) and powerpoint overall quality (max length 10 min)	10 marks

Total: 100 marks